MD. MOAZZEM HOSSAIN MAJUMDER

Dhaka, Bangladesh | +880 1770-303064 | moazzemhossain1510@gmail.com
linkedin.com/in/moazzem-hossain-majumder | github.com/moazzem-hossain-majumder | Portfolio

PROFESSIONAL SUMMARY

Last Semester Computer Science student at BRAC University (CGPA 3.51) seeking AI Engineer Intern and Data Analyst Intern roles. Builds and deploys GenAI applications and agentic workflows (LangChain, LangGraph), multimodal deep learning in PyTorch, and Power BI dashboards with DAX. Holds the IBM Generative AI Engineering and IBM Data Analyst Professional Certificates.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
Generative AI & APIs: LLM APIs (Gemini, Groq, OpenRouter), image generation (FLUX), Whisper, text-to-speech, Hugging Face, prompt engineering, RAG (Chroma), LangChain, LangGraph, AI agents, tool calling, local LLMs (Ollama)
Web & Automation: REST APIs, FastAPI, Flask, Gradio, Streamlit, Pydantic, workflow automation, pytest, GitHub Actions CI, Render deployment
Data Analysis & ML: Pandas, NumPy, data cleaning, EDA, Power BI, DAX, Excel, dashboards, KPI reporting, PyTorch, model evaluation
Tools: Git, GitHub, Jupyter Notebook, LaTeX

PROJECTS

Podcast Companion | *Python, FastAPI, Gemini, Groq, Whisper, Pydantic* | GitHub · Live Demo

- Built and deployed a web app that turns a podcast transcript (text, SRT, VTT) or audio (Whisper) into an overview, timestamped chapters, a quiz and an "ask the episode" Q&A, with Gemini and Groq and automatic model fallback.
- Grounded every output: segment-numbered citations, quiz quotes verified verbatim by the server (unverifiable questions dropped or re-asked), timestamps taken from Whisper and subtitles, and refusal of questions the episode does not cover.
- Benchmarked Gemini vs Groq on latency and validation re-asks; hardened with magic-byte audio checks, rate limiting and CSP; covered by 135 pytest tests.

Quillframe | *Python, FastAPI, Gemini, Groq, OpenRouter, FLUX, edge-tts* | GitHub

- Built a multi-model creative AI workbench that sends one brand brief to Gemini, Groq and OpenRouter in parallel, builds a campaign (concept, slogans, social captions) with model fallback, and generates FLUX images and Bangla-capable voiceovers.
- Logged latency and success rate per tool into an auto-generated experiment report; shipped a web UI, CLI and REST API with schema validation, retries and CI-run pytest tests.

Operations AI Agent | *LangChain, LangGraph, Ollama, SQLite, Chroma, Streamlit* | GitHub

- Built an agentic assistant that chooses among four tools (SQL, policy-document search, statistics) and chains them for multi-step questions, with per-session memory; runs fully locally via Ollama with no API key.
- Cleaned procurement data with Pandas and NumPy, added z-score anomaly flagging and read-only SQL guardrails, and tested with 31 offline tests plus an 8-question evaluation harness for accuracy and tool routing.

Telecom Customer Churn Analytics Dashboard | *Python, Power BI, DAX* | GitHub

- Generated a synthetic 5,000-customer telecom dataset with Python (Pandas, NumPy) and built an interactive Power BI dashboard with slicers, DAX KPIs (churn rate, ARPU, revenue loss) and a division-by-tenure churn heatmap.
- Segmented retained customers into High, Medium and Low churn risk with DAX, surfacing 168 high-risk customers and BDT 77,750 in monthly recharge at risk, and found app users churn at 13.9% vs 25.5% for non-users.

Multimodal V2V Beam Prediction (Manuscript in Preparation) | *PyTorch, DeepSense 6G* | GitHub

- Developed a multimodal pipeline fusing GPS and RGB camera data (ResNet-18, BiGRU, Transformer fusion) to predict the beampower profile of 256-beam 60 GHz vehicle-to-vehicle links on DeepSense 6G Scenario 36.
- Added an online adaptive conformal risk controller that cut beam probing by 93% at a 7.5% miss rate (seed 42).

EDUCATION

Bachelor of Science in Computer Science | **BRAC University** (Jan 2023 – Jan 2027) | CGPA 3.51/4.00 (till 11th semester)
Higher Secondary Certificate (HSC) | *Adamjee Cantonment College, Dhaka* (2021) | Science, GPA 5.00/5.00
Secondary School Certificate (SSC) | *Adamjee Cantonment Public School, Dhaka* (2019) | Science, GPA 5.00/5.00, Scholarship

CERTIFICATIONS

- IBM Generative AI Engineering Professional Certificate, Coursera | Credential
- AI Engineering with LangChain Track, DataCamp | Credential
- CS50P: CS50's Introduction to Programming with Python, Harvard University | Credential
- IBM Data Analyst Professional Certificate, Coursera | Credential